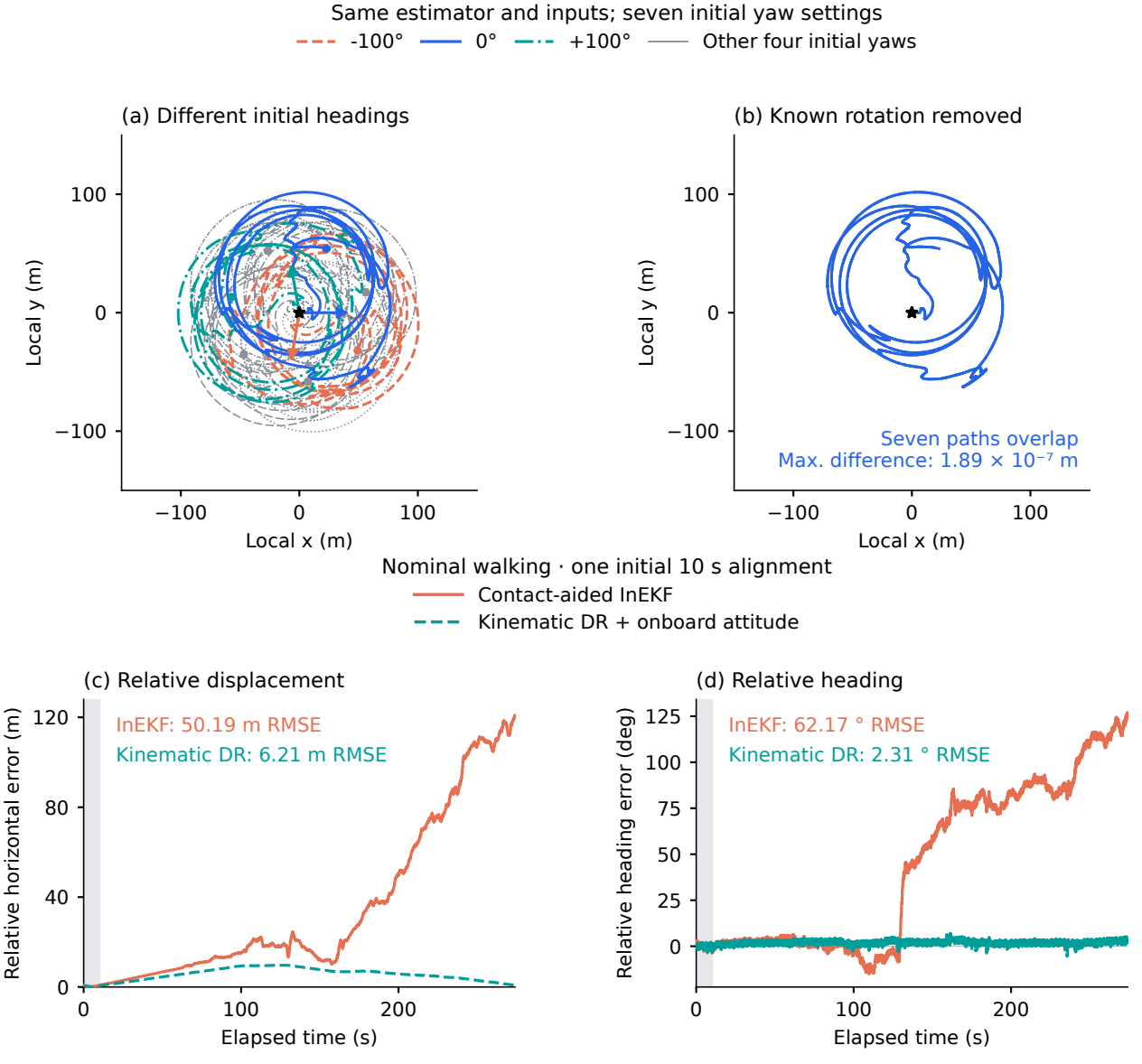

Same estimator and inputs; seven initial yaw settings
-100°
0°
+100°
Other four initial yaws
(a) Different initial headings
(b) Known rotation removed
Local y (m)
Local x (m)
100
0
−100
Seven paths overlap
Max. difference: 1.89 × 10⁻⁷ m
Nominal walking · one initial 10 s alignment
Contact-aided InEKF
Kinematic DR + onboard attitude
(c) Relative displacement
(d) Relative heading
Relative horizontal error (m)
Relative heading error (deg)
Elapsed time (s)
InEKF: 50.19 m RMSE
Kinematic DR: 6.21 m RMSE
InEKF: 62.17 ° RMSE
Kinematic DR: 2.31 ° RMSE
0
40
80
120
25
50
75
125
100
200